TECHNICAL MEMORANDUM

BBMI NCAA Basketball Model

TO:	BBMI Sports Analytics — Internal Technical Record
FROM:	BBMI Quantitative Modeling
DATE:	April 1, 2026
RE:	NCAA Basketball Model — Technical Memorandum
MODEL VERSION:	Production (Progressive Stretch, 11-Feature Ensemble)
VALIDATION PERIOD:	2023–2026 NCAA Seasons (3-Season Walk-Forward)

This memorandum contains proprietary modeling methodology and validation results for the BBMI NCAA Basketball prediction system. Results are based on walk-forward out-of-sample testing. Past performance does not guarantee future results.

1. Executive Summary

The BBMI NCAA Basketball model generates spread predictions, over/under totals, and win probabilities for all Division I men's basketball games. The model uses a percentile-rank ensemble of 11 statistical features, calibrated through a progressive line stretch formula that converts raw BBMI score differentials into point spread projections.

The model's edge derives from its architecture — the progressive stretch formula and KenPom efficiency baseline — rather than from weight optimization. A 3-season actuarial audit with permutation testing confirmed that the optimization surface is flat: the top 20 weight combinations span only 0.23 percentage points (51.24% to 51.47% ATS out-of-sample).

Product	Performance	Sample	Status
ATS Spread (all picks)	~52–53% ATS	Multi-season	Validated
ATS Spread (edge ≥ 5 pts)	56–58% ATS	Filtered subset	Primary product
Over/Under	~52–54% ATS	Multi-season	Validated

2. Model Architecture

The model pipeline executes the following sequential steps:

```
Step 1    Fetch team statistics (ESPN API, KenPom)
Step 2    Calculate recency-weighted stats per team
Step 3    Percentile rank each stat (1–50 scale)
Step 4    Weighted aggregate → BBMI composite score
Step 5    Apply ranking cap (10–20 spots/day max improvement)
Step 6    Calculate residual-based home court advantage per team
Step 7    Raw line = (away_BBMI − home_BBMI + HCA) × multiplier
Step 8    Progressive stretch (calibrated non-linear transformation)
Step 9    Snap to nearest 0.5 (never whole number)
Step 10   Over/Under via KenPom efficiency × pace × modifiers
Step 11   Win probability via Normal CDF
Step 12   Edge calculation (BBMI line vs Vegas line)
```

Design Principles

- Percentile-rank ensemble. Each statistic is ranked on a 1–50 scale across all 351+ D-I teams, then weighted and aggregated. This normalizes different stat scales and reduces sensitivity to outliers.
- Progressive stretch, not linear. A raw BBMI score differential of 2 points does not translate to a 2-point spread. The progressive stretch amplifies small edges more aggressively than large ones, reflecting the non-linear relationship between quality differential and margin of victory.
- Residual-based HCA. Each team's home court advantage is computed from the residual between actual margin and model projection, separately for home and away games. This captures venue-specific crowd effects rather than applying a single national average.
- Ranking cap. Teams cannot improve more than 10–20 ranking spots per day, preventing a single upset win from causing a volatile ranking jump. This stabilizes the model while allowing gradual assessment updates.

3. Input Data and Calibration

3.1 Statistical Features

The model uses 11 features, each percentile-ranked and weighted:

Feature	Weight	Direction	Description
fg_diff	0.156	Higher is better	Field goal percentage differential
adjD	0.139	Lower is better	KenPom adjusted defensive efficiency
turnovers	0.123	Higher is better	Turnover rate differential
ft_pct	0.089	Higher is better	Free throw percentage
rebounds	0.089	Higher is better	Rebound rate differential
assists	0.089	Higher is better	Assists per game
quality_wins	0.084	Higher is better	Schedule-adjusted win value (tier-weighted)
adjO	0.083	Higher is better	KenPom adjusted offensive efficiency
ft_rate_diff	0.080	Higher is better	Free throw rate differential
three_pt_diff	0.034	Higher is better	Three-point percentage differential
last10	0.034	Higher is better	Win-loss record in last 10 games
Total: 1.000			

3.2 KenPom Efficiency Metrics

KenPom's adjusted efficiency ratings measure points scored (adjO) and allowed (adjD) per 100 possessions, adjusted for opponent strength. These are the most predictive single inputs available for college basketball.

- adjO (adjusted offensive efficiency): Higher = better offense
- adjD (adjusted defensive efficiency): Lower = better defense
- adjTempo (adjusted tempo): Possessions per 40 minutes

Collinearity finding: adjO and adjD exhibit near-perfect collinearity ($r = 0.947$ in both 2023–24 and 2024–25). Collapsing to a single adjNetEff = adjO − adjD is out-of-sample neutral (52.2% vs 52.3% ATS). The model retains separate features for now; architectural cleanup is flagged as a deferred item.

3.3 Quality Wins

Games are valued based on the opponent's rank, with bonuses for beating highly ranked teams and penalties for losing to weaker opponents. See Appendix B for the full tier table.

3.4 Percentile Ranking

Each stat is ranked across all D-I teams on a 1–50 scale. The weighted aggregate is then multiplied by 0.8 to produce the final BBMI composite score. A lower-bound adjustment for quality wins compresses the scale to 80%, penalizing teams with weak schedules.

4. Spread Projection Engine

4.1 Raw Line Formula

```
raw_line = (away_BBMI − home_BBMI + HCA) × BBMI_LINE_MULTIPLIER
```

Where:

- BBMI_LINE_MULTIPLIER = 1.42 (OLS-calibrated, March 2026)
- HCA = 3.5 points (baseline; team-specific adjustments via residual method)

4.2 Progressive Stretch

The raw line is transformed through a non-linear function that amplifies small differentials and compresses extreme ones:

```
progressive_line(raw) = sign(raw) × (1.0 × |raw| + 2.0 × √|raw|)
```

Raw Line (pts)	Progressive Output	Effective Multiplier
2.0	4.83	2.4×
5.0	9.47	1.9×
10.0	16.32	1.6×
15.0	22.75	1.5×
20.0	28.94	1.4×

This captures the empirical reality that a team with a slight quality edge wins by a wider margin than a linear model would predict, while extreme quality gaps do not produce proportionally extreme blowouts.

4.3 Home Court Advantage

Baseline HCA: 3.5 points (national average)

Team-specific HCA is computed via the residual method:

```
true_hca       = (home_residual − away_residual) / 2
adjusted_hca   = true_hca × 0.4       (stability scale-down)
capped_hca     = min(adjusted_hca, 7.0)
final_hca      = max(−capped_hca, 0.0)   (never favors away team)
```

4.4 Line Snapping

The final spread is rounded to the nearest 0.5 and never allowed to be a whole number (to prevent pushes on the half-point):

```
if bbmi_line == int(bbmi_line): bbmi_line ± 0.5
```

4.5 Worked Example

Duke (home, BBMI score 42.5) vs North Carolina (away, BBMI score 38.0). Duke team-specific HCA: 4.2 points.

```
raw_line     = (38.0 − 42.5 + 4.2) × 1.42 = –0.426
progressive  = −(0.426 + 2.0 × √0.426) = −(0.426 + 1.306) = −1.73
snapped      = −1.5  (Duke −1.5)
```

Duke is projected as a 1.5-point favorite. If Vegas has Duke −4.5, the edge is 3.0 points toward North Carolina.

5. Over/Under Total Projection

5.1 Pace Projection

```
raw_pace        = (tempo_home + tempo_away) / 2.0
projected_poss  = raw_pace × 0.75 + AVG_TEMPO × 0.25
```

Where AVG_TEMPO = 67.4 possessions (league average, dynamically computed). The 25% regression toward the league mean dampens extreme pace projections.

5.2 Efficiency Interaction

```
home_pts_per_100 = adjO_home × (adjD_away / AVG_ADJ_EFFICIENCY)
away_pts_per_100 = adjO_away × (adjD_home / AVG_ADJ_EFFICIENCY)
```

Where AVG_ADJ_EFFICIENCY = 109.3. This cross-references each team's offensive efficiency against the opponent's defensive efficiency.

5.3 Scale to Game Points

```
home_pts = home_pts_per_100 × projected_poss / 100.0
away_pts = away_pts_per_100 × projected_poss / 100.0
```

5.4 Adjustments Applied

Adjustment	Formula	Notes
HCA total bump	home_pts += abs(hca) × 0.25	Home teams score slightly more
Recent form	±0.25 pts × (last10_wins − 5.0)	Each win above .500 adds 0.25 pts
Neutral site	−2.5 pts total	Tournament/neutral games score lower
FG% / 3PT% modifiers	Disabled (weight = 0.0)	Too noisy game-to-game

6. Win Probability

Win probability is computed using a Normal distribution CDF:

```
Z             = bbmi_line / (STD_DEV × √2)
P(home wins)  = 1 − erfc(Z) / 2
```

Where STD_DEV = 10.75 points, calibrated to minimize Brier score across the walk-forward sample.

Example: BBMI line = −6.5 (home favored by 6.5 points).

```
Z            = −6.5 / (10.75 × 1.414) = −6.5 / 15.20 = −0.428
P(home wins) = 1 − erfc(−0.428)/2 ≈ 0.666 = 66.6%
```

7. Product Definitions and Examples

7.1 ATS Spread Pick

Signal: BBMI's projected spread differs from the Vegas spread by a meaningful amount.

```
edge           = |bbmi_line − vegas_line|
pick direction: if bbmi_line > vegas_line → pick away; if < → pick home
```

Example: BBMI projects Kansas −3.5. Vegas has Kansas −8.5. Edge = 5.0 points. Pick: Kansas opponent (cover +8.5).

7.2 Edge Classification

Edge	Classification	Description
≥ 5.0 pts	**High edge**	Primary product — strongest signal
3.0–5.0 pts	**Medium edge**	Secondary — included with lower conviction
0.5–3.0 pts	**Low edge**	Marginal — included for completeness
< 0.5 pts	**No edge**	Filtered out

7.3 Over/Under Pick

```
ou_edge = bbmi_total − vegas_total
if ou_edge > threshold  →  OVER
if ou_edge < −threshold →  UNDER
```

8. Walk-Forward Validation Protocol

8.1 Methodology

The model was validated across three seasons using walk-forward methodology, where no future information enters any projection. KenPom snapshots are locked per walk-forward year.

- 2023–24: Training data
- 2024–25: Out-of-sample test (trained on 2023–24)
- 2025–26: Live season with rolling updates

8.2 Permutation Test

A permutation test was conducted to verify that the model's performance is not an artifact of optimization:

- Method: Shuffle game outcomes randomly, re-run the full weight optimization.
- Result: Shuffled data produces the same best ATS (51.47%) as real data.
- Conclusion: Weight optimization finds patterns in search space geometry, not in underlying signal. The model's edge derives from its architecture (progressive stretch, KenPom baseline), not from weight tuning.

8.3 Seasonal Windows

Window	Months	Characteristics
Early season	Nov–Dec	Highest uncertainty; heaviest reliance on preseason ratings
Mid-season	Jan–Feb	Sample accumulation; weight recalibration
Late season	Mar–Apr	Conference tournaments; NCAA tournament

9. Validated Results

9.1 Spread ATS Summary

Metric	Value
Overall ATS (3-season)	**~52–53%**
High-edge ATS (≥ 5 pts)	**56–58%**
Progressive stretch calibration (in-sample)	**75.2% on 2,231 picks**
Optimization surface span (top 20 combos)	**0.23 pp**
Equal weights baseline (OOS)	**51.1%**

9.2 Seasonal Stability

Season Window	ATS	Notes

2023–24 late season	41.6%	Genuine outlier
2024–25 tournament	51.8%	Adequate
2025–26 tournament	53.8%	Above break-even
2025–26 March production	**52.7%**	419 games — above break-even

9.3 Calibration History

Parameter	Original	Current	Calibration Date
BBMI_LINE_MULTIPLIER	Variable	**1.42**	2026-03-03 (OLS fit)
STD_DEV	9.0	**10.75**	2026-03-14 (Brier score)
Progressive Stretch D	N/A	**2.0**	2026-03-20 (75.2% ATS)

10. Components Investigated and Excluded

10.1 Tournament Stretch Dampener

Hypothesis: The progressive stretch formula over-amplifies edges in March Madness, causing systematic favorites bias in tournament games.

Testing: Tested D = 0.0 (pure linear) through D = 2.0 (production) across 938 tournament games over 3 seasons.

Result: ATS range 50.2% to 50.6% — completely flat. No dampener value improves tournament performance. Large Elite 8 edges are individual game noise, not systematic miscalibration.

Conclusion: Permanently excluded. Production D = 2.0 retained.

10.2 adjO/adjD Collapse to Net Efficiency

Hypothesis: Collapsing highly collinear adjO and adjD (r = 0.947) to a single net efficiency metric would simplify the model without losing performance.

Result: Out-of-sample neutral (52.2% vs 52.3% ATS). An architectural simplification, not a performance improvement.

Status: Deferred. Will activate only after production snapshot data confirms collinearity in live model.

10.3 ft_rate_diff Weight Reduction

Finding: ft_rate_diff was overweighted at 0.123. Stable at 0.06–0.08 across all validation cuts: 3-season, 2-season, every window, constrained and unconstrained.

Action: Cut to 0.080 and shipped. The only weight change from the full audit that survived every check.

10.4 Turnovers Weight Increase

Finding: Constrained grid search placed turnovers at 0.130 (vs current 0.123) — directionally consistent but within grid step size (0.03). Signal unstable: shifts from 0.130 (3-season) to 0.100 (2-season excluding 2023–24).

Status: Deferred until a clean production snapshot dataset exists.

10.5 FG% and 3PT% Modifiers in Totals

Status: Both weights zeroed in the O/U model. Game-to-game shooting percentages are too noisy to improve total projections. KenPom efficiency already captures sustained shooting ability.

11. Open Items and Forward-Looking Agenda

11.1 Offseason Roadmap (2026)

Item	Target	Description
BoundScraper git + process controls	**Complete**	Version control for pipeline changes
fakeWin test	August 2026	Verify no data leakage in win/loss encoding
Snapshot validation	Mid-offseason	Verify feature snapshots match point-in-time enforcement
adjO/adjD collapse	Mid-offseason	If collinearity confirmed in production data, simplify to net efficiency
Phase 2 walk-forward	2-season project	Full multi-season walk-forward with production snapshots

11.2 Monitoring Metrics

- Weekly ATS by edge bucket
- Monthly ATS by seasonal window (Early / Mid / Late)
- Conference tournament run-up week (2/23–3/1 soft spot monitoring)
- O/U ATS with pace-adjusted analysis

Appendix A — Statistical Feature Weights

Stat	Weight	Direction	Data Source
fg_diff	0.156	Higher = better	ESPN box scores
adjD	0.139	Lower = better	KenPom
turnovers	0.123	Higher = better	ESPN box scores
ft_pct	0.089	Higher = better	ESPN box scores
rebounds	0.089	Higher = better	ESPN box scores
assists	0.089	Higher = better	ESPN box scores
quality_wins	0.084	Higher = better	Computed from results + opponent rank
adjO	0.083	Higher = better	KenPom
ft_rate_diff	0.080	Higher = better	ESPN box scores
three_pt_diff	0.034	Higher = better	ESPN box scores
last10	0.034	Higher = better	ESPN results

Appendix B — Quality Wins Tiers

Opponent Rank	Win Bonus	Close Loss (≤5 pts)	Loss Penalty
1–15 (Tier 1)	+15.0	+3.0	−2.5
16–35 (Tier 2)	+7.5	+1.0	−4.0
36–100 (Tier 3)	+2.0	—	−8.0
101+ (Tier 4)	—	—	−5.0

Close losses against elite opponents are valued positively — a 3-point road loss to a top-15 team demonstrates competitive quality.

Appendix C — Glossary

Term	Definition
BBMI Score	Composite percentile-ranked team rating (higher = better)
Progressive Stretch	Non-linear transformation converting raw score differentials to point spreads; amplifies small edges, compresses large ones
adjO / adjD	KenPom adjusted offensive/defensive efficiency (points per 100 possessions vs average opponent)

adjTempo	KenPom adjusted tempo (possessions per 40 minutes)
Ranking Cap	Maximum daily ranking improvement (10 spots if ranked ≤75, 20 spots if >75)
Residual HCA	Team-specific home court advantage derived from the gap between actual and projected margins at home vs away
Permutation Test	Statistical test that shuffles outcomes to determine if optimization results exceed chance
STD_DEV	Standard deviation of score margin distribution (10.75 pts), used for win probability calculation

This memorandum describes the BBMI NCAA Basketball model as of April 1, 2026. Model parameters and validation results are subject to revision based on ongoing monitoring and the 2026 offseason review.